RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science Degree in Applied Sciences
Third Year - Semester II Examination – February / April 2025

COM 3401 – DATA COMMUNICATION AND NETWORKING

Time: Three (03) hours

- Answer **ALL** questions.
- This paper contains five (05) questions in three (03) pages.

1. a) Consider a communication between a personal computer and a server over a public telephone network.

i. Suggest the key elements that should be present in a simple communication model that can be used to represent this communication.

ii. Draw a suitable block diagram to illustrate the above model.

iii. Briefly state the function of each of the elements.

(06 marks)

b) Explain the following tasks in a data communication system:

i. Transmission system utilization.

ii. Exchange management.

(04 marks)

c) Distinguish between the following terms in data transmission:

i. Point-to-point and multi-point link.

ii. Simplex and full-duplex transmission.

(04 marks)

d) Briefly explain the following advantages of digital data transmission over analog data transmission:

i. Technology advantages.

ii. Capacity utilization.

iii. Integration.

(06 marks)

2. a) Briefly explain how an electromagnetic signal is represented in the frequency domain and what frequency-related information can be obtained from that representation.
(05 marks)

b) Explain the need for modulation techniques and the role of the carrier signal in data transmission.
(04 marks)

c) How does the carrier signal change during the ASK process? Draw a suitable diagram to illustrate the signal elements of the modulated signal for the bit stream 00110100.
(04 marks)

d) What is the function of twisting in a twisted-pair cable and why is the twisted-pair shielded?
(04 marks)

e) Indicate three (03) significant differences between broadcast radio and microwave.
(03 marks)

3. a) Briefly explain the key differences and similarities between bridges and routers.
(04 marks)

b) List four (04) characteristics that can be used to classify computer networks.
(04 marks)

c) Discuss the need for, or lack of need for, an OSI Layer 3 in a broadcast network.
(04 marks)

d) Which layers of the TCP/IP model are implemented in routers? Explain your answer.
(04 marks)

e) "The application layer in the Internet includes many predefined protocols to support various user applications." List four (4) of them along with their applications.
(04 marks)

4. a) Explain the Bit Stuffing method in framing. Show how this method is applied to the bit stream 01111110111110111111111110.
(04 marks)

b) In CRC, if the dataword is 10 bits and the codeword is 14 bits:

i. How many 0s need to be added to the dataword to make the dividend?

ii. What is the size of the FCS?

iii. What is the size of the divisor?

iv. Suggest a suitable divisor.

(04 marks)

c) Explain how synchronous TDM works.
(04 marks)

d) Why is statistical TDM more efficient than synchronous TDM?
(04 marks)

e) Assume there are five (05) channels, each with a bandwidth of 10 kHz and a guard band of 500 Hz, to be multiplexed using FDM. Calculate the total required bandwidth.
(04 marks)

5. a) Which transport layer protocol, TCP or UDP would be more appropriate for text messaging? Justify your decision.

(04 marks)

b) Find the network identifier and host identifier of each of the following classful IP addresses:

i. 156.43.24.32

ii. 199.60.52.121

(04 marks)

c) Consider the block of addresses containing the IP address 20.16.4.18/28:

i. Write the network mask of the IP address in dotted decimal notation.

ii. Find the number of addresses in the block.

iii. What are the first and last addresses in this block?

(06 marks)

d) Suppose a block of IP addresses starting with 34.17.4.128/25 is granted to an organization. The organization needs to divide this address block into three (03) subnets, with 32, 64, and 32 addresses in each subnet. Find the subnet mask and subnet address for each subnet.

(06 marks)

--- End ---